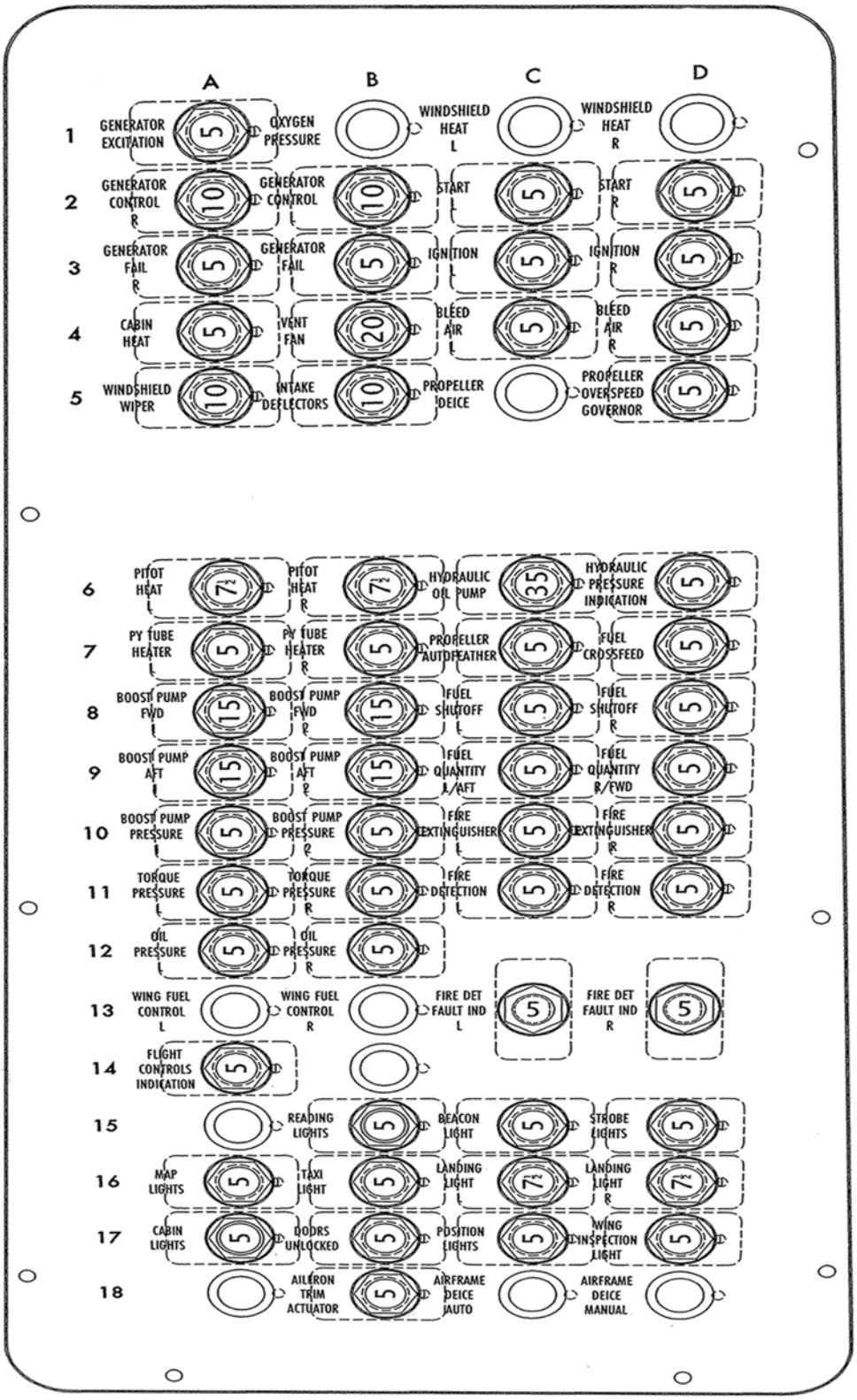
DC Electrical Load Distribution – General Data

1. General

DC power distribution is by means of a multiple bus system consisting of left and right 28 Volt DC buses, and main battery, battery/external power. The left generator is connected to the left DC bus and the right generator to the right DC bus by LEFT and RIGHT GENERATOR switches marked OFF–ON–RESET, together with a POWER SOURCE switch and MASTER switch. When Mod 6/2314 is installed, the POWER SOURCE and MASTER switches are deleted, and replaced with BATTERY and EXTERNAL switches which independently select DC voltages from the BATTERY or EXTERNAL sources. All buses can be supplied from one generator or the main battery, through a bus-tie relay operated when a BUS TIE switch is set to NORMAL. The switches are located on the overhead console panels. The main battery bus, connected directly to the main battery positive terminal, is connected through a battery relay to the battery/external power bus, then through a reverse current circuit breaker to the left DC bus. If the BUS TIE switch is set to OPEN, the right DC bus is isolated and is only powered from the right generator. Power is distributed to the left and right DC buses, through bus feed circuit breakers in the power distribution and generator control box and through circuit breakers on the main overhead and avionics distribution panels added for Mod 6/1274, to supply electrical circuits through individual system circuit breakers (refer to Figure 1), mounted on panels in the flight compartment, the main panels on the bulkhead to the left of the pilot, and avionics panels on the center pylon (refer to Figure 1). Circuits from the main battery bus operate cabin entrance and baggage compartment lights. (Refer to 33-10-00, Flight Compartment Lights (Mod 6/2028) – General Data).

Six bus feed, 50 ampere circuit breakers (left DC CB5, CB7, CB9 and right DC CB4, CB6, CB8) are installed in the power distribution and generator control box (24-60-11, Power Distribution and Generator Control Box (Mod 6/2028) – General Data, Figure 1) to supply six 50-ampere circuit breakers (CB5A, CB7A, CB9A, CB4A, CB6A and CB8A) on the main power distribution circuit breaker panel (refer to Figure 3). Six bus feed, 30-ampere circuit breakers (left DC CB13, CB15, CB17 and right DC CB12, CB14, CB16) are installed in the power distribution and generator control box to six 30-ampere circuit breakers (CB13A. CB15A, CB17A, CB12A. CB14A and CB16A) on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).



Circuit Breaker Panel (Typical)
Figure 1

Table 1 Main Circuit Breaker Panel CBP-A

Left Bus		Amperage	SOO
GENERATOR EXCITATION	CB101	5	
GENERATOR CONTROL R	CB103	10	
GENERATOR FAIL R	CB105	5	
CABIN HEAT	CB107	5	
WINDSHIELD WIPER	CB109	10	
PITOT HEAT	CB111	7.5	
PY TUBE HEATER L	CB113	5	
BOOST PUMP FWD 1	CB115	15	PRE MOD 6/2299
		10	POST MOD 6/2299
BOOST PUMP AFT 1	CB117	15	PRE MOD 6/2299
		10	POST MOD 6/2299
BOOST PUMP PRESSURE 1	CB119	5	
TORQUE PRESSURE L	CB121	5	
OIL PRESSURE L	CB123	5	
WING FUEL CONTROL L	CB125	7.5	6247
FLIGHT CONTROL INDICATION	CB127	5	
RESERVED	CB129		
MAP LIGHTS	CB131	5	
CABIN LIGHTS	CB133	5	
RESERVED	CB135		
WINDSHIELD HEAT L	CB137	5	6187
START L	CB139	5	
IGNITION L	CB141	5	
BLEED AIR L	CB143	5	
PROPELLER DEICE	CB145	20	6202
HYDRAULIC OIL PUMP	CB147	35	
PROPELLER AUTOFEATHER	CB149	5	
FUEL SHUTOFF L	CB151	5	
FUEL QUANTITY L/AFT	CB153	5	
FIRE EXTINGUISHER L	CB155	5	
FIRE DETECTION L	CB157	5	
FIRE DET FAULT IND L	CB161	5	
BEACON LIGHT	CB165	5	

Table 1 Main Circuit Breaker Panel CBP-A (continued)

Left Bus		Amperage	SOO
LANDING LIGHT L	CB167	7.5	PRE MOD 6/2285
		5	POST MOD 6/2285
POSITION LIGHTS	CB169	5	
AIRFRAME DEICE AUTO	CB171	5	6004

Right Bus		Amperage	SOO
RESERVED	CB100		
GENERATOR CONTROL L	CB102	10	
GENERATOR FAIL L	CB104	5	
VENT FAN	CB106	20	
INTAKE DEFLECTORS	CB108	10	
PITOT HEAT R	CB110	7.5	
PY TUBE HEATER R	CB112	5	
BOOST PUMP FWD 2	CB114	15	PRE MOD 6/2299
		10	POST MOD 6/2299
BOOST PUMP AFT 2	CB116	15	PRE MOD 6/2299
		10	POST MOD 6/2299
BOOST PUMP PRESSURE 2	CB118	5	
TORQUE PRESSURE R	CB120	5	
OIL PRESSURE R	CB122	5	6247
WING FUEL CONTROL R	CB124	7.5	
TAXI LIGHT R	CB126	5	6441, 6442
READING LIGHTS	CB128	5	
TAXI LIGHT	CB130	5	
TAXI LIGHT L	CB130	5	6441, 6442
DOORS UNLOCKED	CB132	5	
AILERON TRIM ACTUATOR	CB134	5	
WINDSHIELD HEAT R	CB136	5	6187
START R	CB138	5	
IGNITION R	CB140	5	
BLEED AIR R	CB142	5	
PROPELLER OVERSPEED GOVERNOR	CB144	5	
HYDRAULIC PRESSURE INDICATION	CB146	5	
FUEL CROSSFEED	CB148	5	
FUEL SHUTOFF R	CB150	5	

Right Bus		Amperage	SOO
FUEL QUANTITY R/FWD	CB152	5	
FIRE EXTINGUISHER R	CB154	5	
FIRE DETECTION R	CB156	5	
FIRE DET FAULT IND R	CB160	5	
STROBE LIGHTS	CB164	5	
LANDING LIGHT R	CB166	7.5	PRE MOD 6/2285
		5	POST MOD 6/2285
WING INSPECTION LIGHT	CB168	5	
AIRFRAME DEICE MANUAL	CB170	7.5	6004
CABIN FANS	CB173	15	6378 6379

Table 2 Avionics Circuit Breaker Panel CBP-B

Left Bus		Amperage	SOO
MAU 1A	CB201	20	
MAU 2A	CB203	20	
AURAL ALERT 1	CB205	1	
PILOT PFD	CB207	5	
UPPER MFD	CB209	5	
ADAHRS 1A	CB211	3	
FLT CTRL 1A	CB213	3	
PITCH TRIM	CB215	5	6221
AP SERVOS	CB217	7.5	6221
PILOT PFD CTRL	CB219	3	
PILOT AUDIO	CB221	5	
MMDR 1	CB223	7.5	
XPDR 1	CB225	5	
DME 1	CB227	3	
GPS 1	CB229	3	
CVR	CB231	5	
FDR	CB233	5	
CAUTION LTS NORMAL	CB235	2	
WX RADAR	CB237	5	
MFD CTRL	CB239	3	
DAU 1A	CB241	5	
DAU 2A	CB243	5	

Table 2 Avionics Circuit Breaker Panel CBP-B (continued)

Left Bus		Amperage	SOO
TAWS	CB245	3	
TCAS	CB247	5	
RAD ALT 1	CB249	3	
ELT NAV	CB251	1	
CABIN PA	CB253	5	
ESIS BATTERY	CB255	15	
ESIS INST	CB257	3	
ESIS MAG	CB259	1	
AVIONICS FAN	CB261	1	
RESERVED	CB263		
RESERVED	CB265		
GEAR IND	CB267	5	6001
RESERVED	CB269		
RESERVED	CB271		
AVIONICS STRT BATT CHARGE	CB273	30	POST MOD 6/2315
AVIONICS STRT BATT CTRL	CB275	2	POST MOD 6/2315
PANEL LIGHTS L	CB277	5	
FLOOD LIGHTS	CB279	5	

Right Bus		Amperage	SOO
MAU 1B	CB200	20	
MAU 2B	CB202	20	
AURAL ALERT 2	CB204	1	
COPILOT PFD	CB206	5	
LOWER MFD	CB208	5	
ADAHRS 1B	CB210	3	
FLT CTRL 1B	CB212	3	
CAUTION LTS BYPASS	CB214	2	
RESERVED	CB216		
COPILOT PFD CTRL	CB218	3	
COPILOT AUDIO	CB220	5	
MMDR 2	CB222	7.5	
XPDR 2	CB224	5	
DME 2	CB226	3	
GPS 2	CB228	3	

Right Bus		Amperage	SOO
SATELLITE WX	CB230	5	6211
STORM SCOPE	CB232	3	6215
SAT COM	CB234	3	6298
HF CTRL	CB236	2	6215
ECTM/FOQA	CB238	3	6288
DAU 1B	CB240	5	
DAU 2B	CB242	5	
HEADSETS	CB244	2	
RESERVED	CB246		
RAD ALT 2	CB248	3	6218
RESERVED	CB250		
28V INPUT	CB252	5	
		20	6282
14V INPUT	CB254	7.5	
RESERVED	CB256		
VHF-FM	CB258	3	6296

Table 3 CBP-C Main Circuit Distribution Panel

Left Bus	Amperage	SOO
CB5A	50	
CB7A	50	
CB9A	50	

Right Bus	Amperage	SOO
CB4A	50	
CB6A	50	
CB8A	50	

Table 4 FWD Converter Circuit Breaker Panel

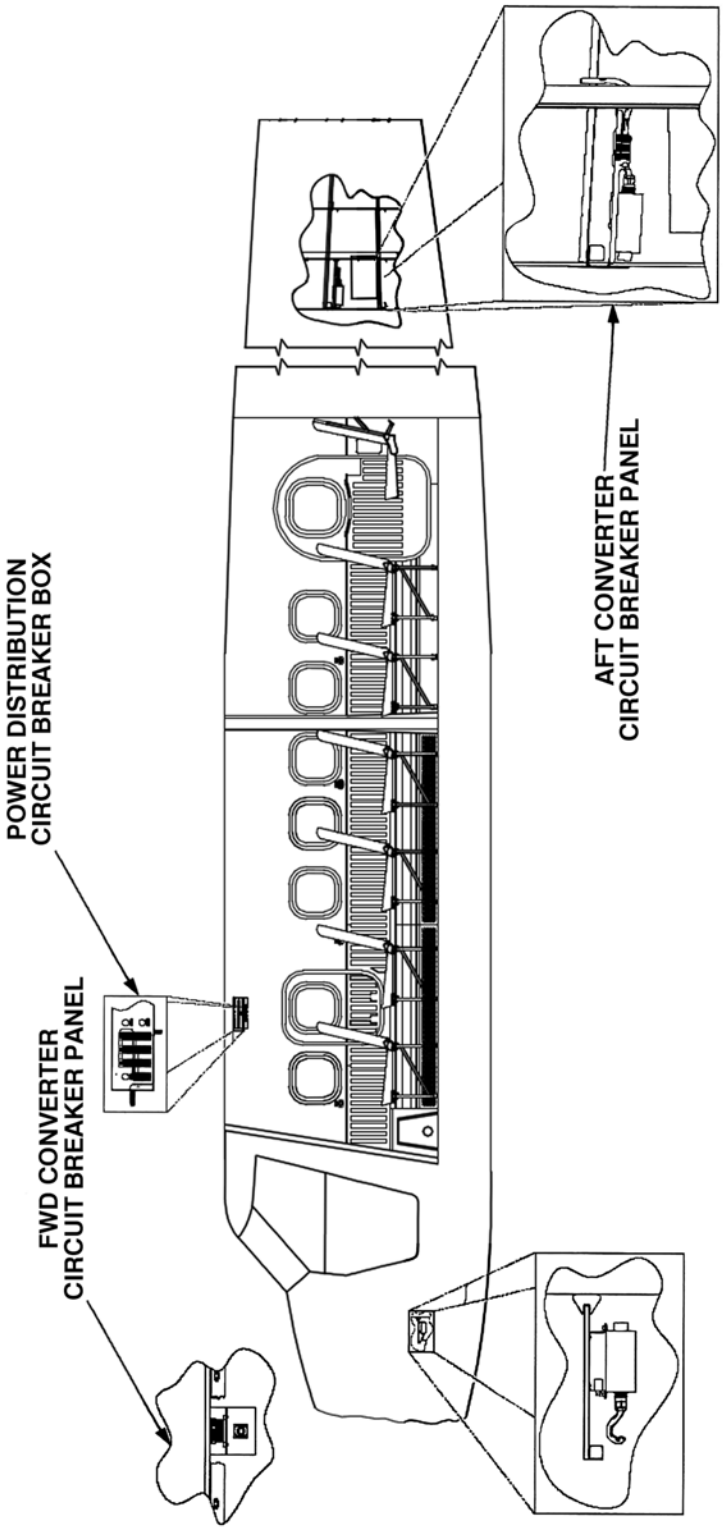
Right Bus		Amperage	SOO
STA 128 14VDC	CB282	7.5	6282
STA 176L 14VDC	CB281	7.5	6282
STA 176R 14VDC	CB280	7.5	6282

Table 5 AFT Converter Circuit Breaker Panel

Right Bus		Amperage	SOO
STA 223	CB286	7.5	6281
STA 258	CB285	7.5	6281
STA 332	CB284	7.5	6281
		7.5	6272

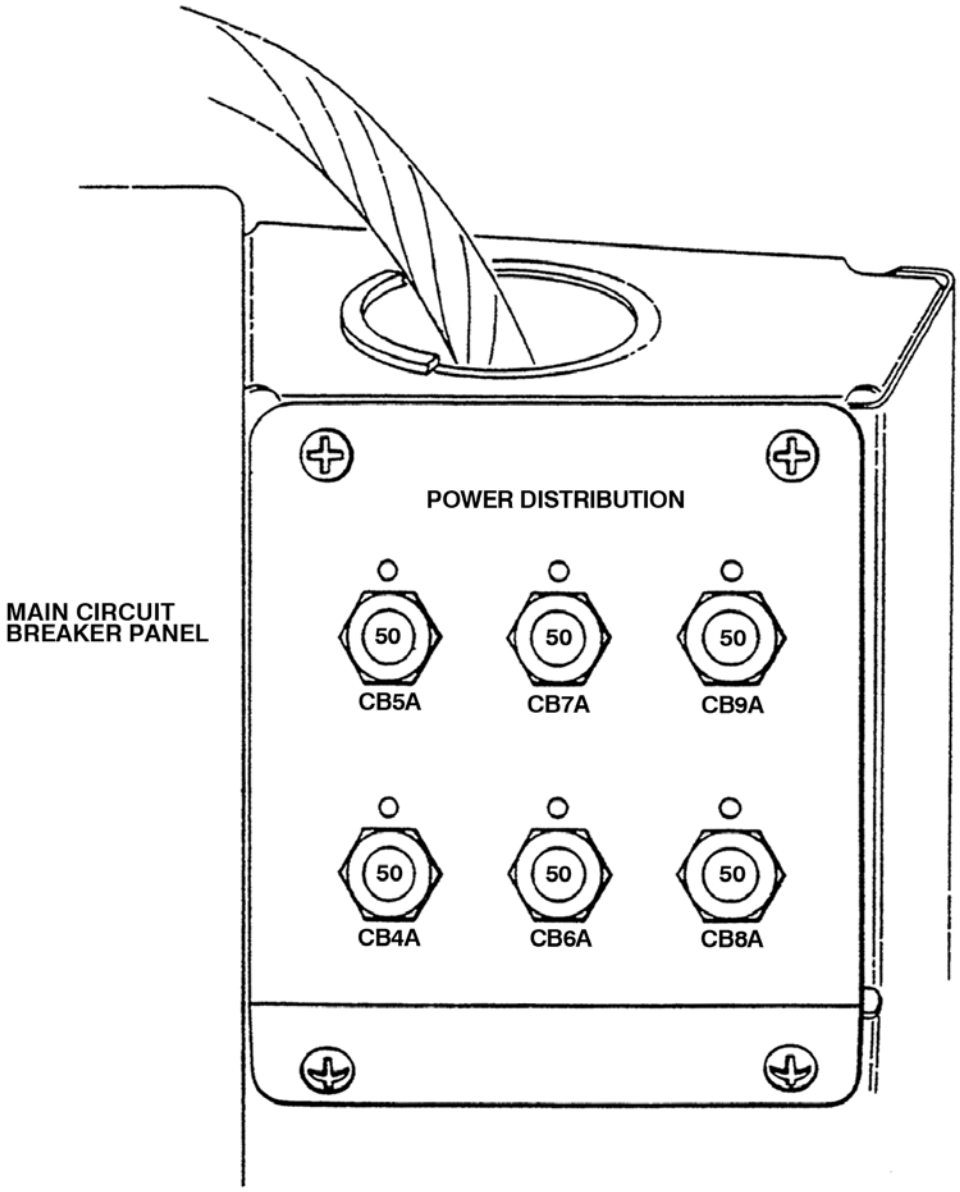
Table 6 Power Distribution Circuit Breaker Box

Right Bus		Amperage	SOO
CBP-E	CB18	5	6272
CBP-E	CB18	15	6281
CBP-E	CB18	5	6283



Cabin Power Converter and Power Distribution Box Locations (S.O.O. 6272, 6281, 6282, 6283)
Figure 2

Print Date: 2025-09-12



Main Circuit Breaker Panel
Figure 3